What I've been building

I've been writing a Python research system that looks for Lorentz-invariant scalars built from differential form field strengths. The case I care about most is a chiral (self-dual) 5-form in ten dimensions. Analytical work suggests there should be on the order of 81 primary invariants; the hard part is producing an explicit list of generators you can actually work with.

Once you have those independent scalars, the most general Lorentz-invariant Lagrangian for the field (no derivatives) is just an arbitrary function of them. In 10D that classification shows up in ModMax-type / TTbar-like deformations of chiral form theories, including ideas related to Type IIB and F5.

I'm not training a neural net. This is closer to computational algebra than deep learning: generate candidate contractions as graphs, turn each into an einsum, evaluate on random chiral tensors with NumPy, then use SVD / numerical rank to see what's independent. Products of lower-order invariants get filtered so the count doesn't inflate from composites.

Around that I built an autonomous runner — presets, checkpoints, a small local dashboard, and a supervisor that keeps the search going overnight. So far it's holding at 78 out of the hypothesized ~81.

In short: a search-and-rank engine for tensor invariants. The math says roughly how many there should be; the code tries to find them explicitly.

Inspired by Elamaran–Ferko–Scarlett, Machine Learning Invariants of Tensors (arXiv:2512.23750). Their paper uses ML language; my implementation so far is numerical linear algebra plus search.